



SWAYAM NPTEL COURSE ON STATISTICAL FOUNDATION FOR BIG DATA ANALYSIS

BY

PROF. ARINDAM BANERJEE

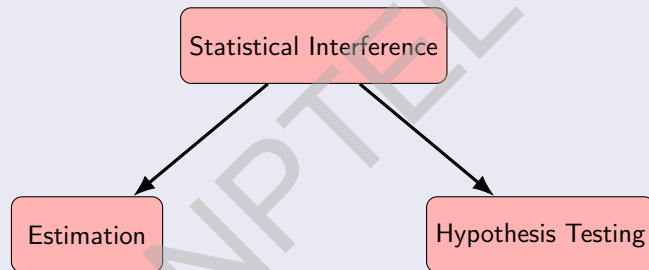
Department of Mathematics
INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

Module 03: Statistical Inference 2

Lecture 01: Interval Estimation 1



Statistical Inference



Estimation

Tries to predict a value related to the population using a sample statistic $T(X_1, \dots, X_n)$.

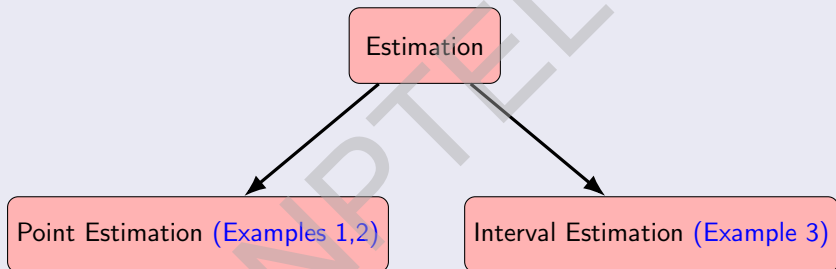
- Population: Follows the distribution of some random variable X .
- Sample: $X_1, \dots, X_n$ such that $F_{X_i} = F_X$ for all i .
- Sample Statistic: A function of sample $T(X_1, \dots, X_n)$.

Estimation (continued)

Example:

- 1 Predict the head probability of a coin using the number of heads in n tosses.
- 2 Predict the mean of a Normal Distribution population by $\frac{X_1 + \dots + X_n}{n}$ where $X_1, \dots, X_n$ is a sample from that.
- 3 Try to predict an interval within the variance of a normal distribution that is very likely to belong using $\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$ where $X_1, \dots, X_n$ is the sample and $\bar{X} = \sum_{i=1}^n X_i$, the sample mean.

Estimation (continued)

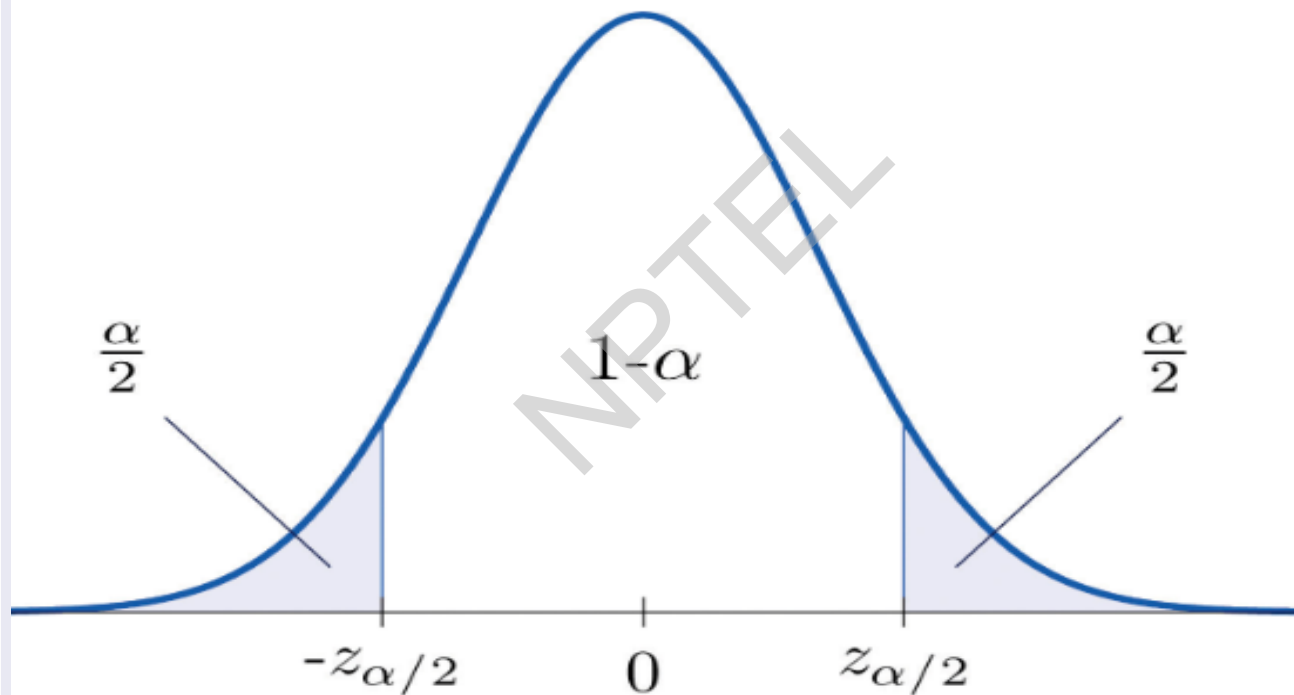


Definition

Let $X_1, \dots, X_n$ be i.i.d sample from some population with parameter θ . Let $0 < \alpha < 1$. Then $100(1 - \alpha)\%$ *confidence interval* is an interval $[c_1(X_1, \dots, X_n), c_2(X_1, \dots, X_n)]$ such that the probability

$$P(\theta \in [c_1(X_1, \dots, X_n), c_2(X_1, \dots, X_n)]) = 1 - \alpha$$

where $c_1(X_1, \dots, X_n) < c_2(X_1, \dots, X_n)$ and $c_1(X_1, \dots, X_n)$ & $c_2(X_1, \dots, X_n)$ are two functions of $X_1, \dots, X_n$.



Useful Results

- **The Weak Law of Large Numbers (WLLN):** Let $X_1, \dots, X_n$ be i.i.d. random variables with a finite expected value μ . Then for any $\epsilon > 0$

$$\lim_{n \rightarrow \infty} P(|\bar{X} - \mu| \geq \epsilon) = 0.$$

where $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$

- Also

$$\begin{aligned} \text{Var}(\bar{X}) &= \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) \\ &= \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) \\ &= \frac{n\sigma^2}{n^2} \\ &= \frac{\sigma^2}{n} \end{aligned}$$

Useful Results

- **Central Limit Theorem (CLT):** Let $X_1, \dots, X_n$ be i.i.d. random variables with expected value $\mu < \infty$ and variance $\sigma^2 < \infty$. Then the random variable

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

is a standard normal variate as $n \rightarrow \infty$.

Useful Results (continued)

- Let $X_1, \dots, X_n$ be i.i.d $\mathcal{N}(\mu, \sigma^2)$ random variables. Also, let S^2 be the sample variance for the random sample. Then the random variable T defined as

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

has t - distribution with $n - 1$ degrees of freedom, i.e., $T \sim t_{n-1}$.
Recall that the pdf of that distribution is

$$\frac{\Gamma(\frac{n+1}{2})}{\sqrt{\pi n} \Gamma(\frac{n}{2})} \left(1 + \frac{x^2}{n}\right)^{-\frac{n+1}{2}}$$

where $n > 0$ is the degrees of freedom and $x \in (-\infty, \infty)$

Useful Results (continued)

- Let $X_1, \dots, X_n$ be i.i.d. $\mathcal{N}(\mu, \sigma^2)$ random variables. Also let S^2 be the sample variance. If

$$Y = \frac{(n-1)S^2}{\sigma^2}$$

has the χ^2 distribution with $(n-1)$ degrees of freedom.

Confidence Interval for Mean When Variance is Known

- Let $X_1, \dots, X_n$ be a random sample from the $\mathcal{N}(\mu, \sigma^2)$ distribution.
- We want to estimate the parameter μ assume that σ^2 is known.
- Construct the random variable

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

- Z follows $\mathcal{N}(0, 1)$ distribution.

Confidence Interval for Mean When Variance is Known (continued)

- For a $100(1 - \alpha)\%$ confidence interval, (where $0 < \alpha < 1$), significance level = $1 - (1 - \alpha) = \alpha$.
For the interval estimation (two tailed) we have,

$$P(-z_{\frac{\alpha}{2}} \leq Z \leq z_{\frac{\alpha}{2}}) = 1 - \alpha$$

$$\text{or, } P\left(-z_{\frac{\alpha}{2}} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq z_{\frac{\alpha}{2}}\right) = 1 - \alpha$$

$$\text{or, } P\left(-z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \leq \bar{X} - \mu \leq z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$

$$\text{or, } P\left(\bar{X} - z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$

- Therefore $\left[\bar{X} - z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}\right]$ is a $100(1 - \alpha)\%$ confidence interval for μ .

Solved Example 1

Question : Suppose the following five numbers were sampled from a normal distribution with a standard deviation 2.5 : 2,3,5,6 and 9. Compute a 95% confidence interval for mean.

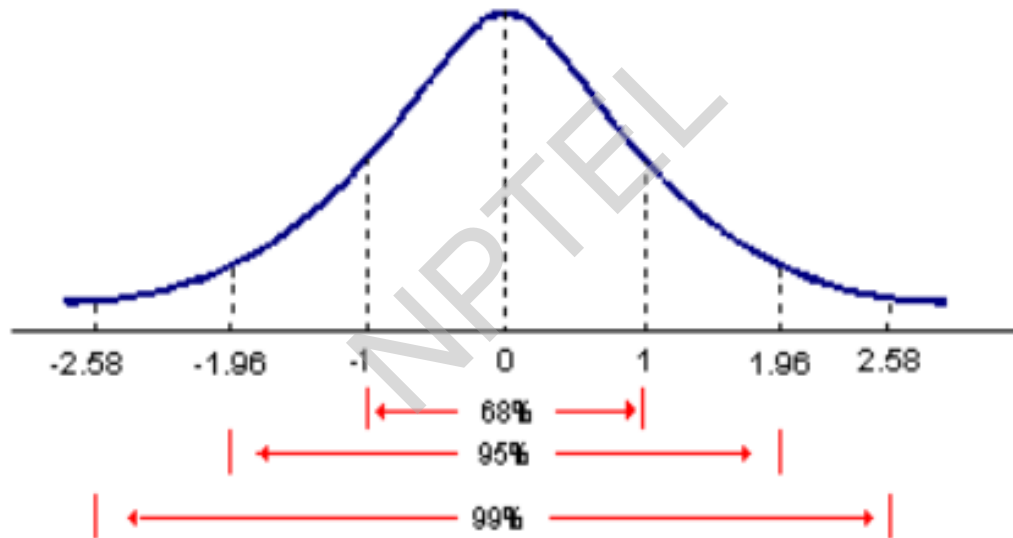
Solution: Here

$$\begin{aligned}\bar{X} &= \frac{2+3+5+6+9}{5} \\ &= 5\end{aligned}$$

and $\sigma = 2.5$. From table $z_{0.025} = 1.96$ Therefore a 95% confidence interval for μ is

$$\begin{aligned}&= \left[\bar{X} - z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \right] \\ &= \left[5 - \frac{1.96 \times 2.5}{\sqrt{5}}, 5 + \frac{1.96 \times 2.5}{\sqrt{5}} \right] \\ &= [2.81, 7.19]\end{aligned}$$

Standard Normal Distribution



Remarks

If $X_1, \dots, X_n$ are random samples from some distribution with mean μ and variance σ^2 . Then for large n , the random variable

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

follows $\mathcal{N}(0, 1)$ distribution where $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ and we can estimate the parameter as before.

Confidence Interval for Mean When Variance is Unknown

- Let $X_1, \dots, X_n$ be a random sample from the $\mathcal{N}(\mu, \sigma^2)$ distribution.
- We want to estimate the parameter μ and assume that σ^2 is unknown.
- Construct the random variable

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

where $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ is the sample variance.

- Then T has t - distribution with $n - 1$ degrees of freedom.

Confidence Interval for Mean When Variance is Unknown (continued)

- For a $100(1 - \alpha)\%$, where $(0 < \alpha < 1)$, confidence interval estimation (two tailed) we have,

$$P(-t_{\frac{\alpha}{2}, n-1} \leq T \leq t_{\frac{\alpha}{2}, n-1}) = 1 - \alpha$$

$$\text{or, } P\left(-t_{\frac{\alpha}{2}, n-1} \leq \frac{\bar{X} - \mu}{S/\sqrt{n}} \leq t_{\frac{\alpha}{2}, n-1}\right) = 1 - \alpha$$

$$\text{or, } P\left(-t_{\frac{\alpha}{2}, n-1} \frac{S}{\sqrt{n}} \leq \bar{X} - \mu \leq t_{\frac{\alpha}{2}, n-1} \frac{S}{\sqrt{n}}\right) = 1 - \alpha$$

$$\text{or, } P\left(\bar{X} - t_{\frac{\alpha}{2}, n-1} \frac{S}{\sqrt{n}} \leq \mu \leq \bar{X} + t_{\frac{\alpha}{2}, n-1} \frac{S}{\sqrt{n}}\right) = 1 - \alpha$$

- Therefore $\left[\bar{X} - t_{\frac{\alpha}{2}, n-1} \frac{S}{\sqrt{n}}, \bar{X} + t_{\frac{\alpha}{2}, n-1} \frac{S}{\sqrt{n}}\right]$ is a $100(1 - \alpha)\%$ confidence interval for μ .

Solved example 2

Question : A farmer weights 10 randomly chosen watermelons from his farm and he obtains the following values (in lbs):

7.72 9.58 12.38 7.77 11.27 8.80 11.10 7.80 10.17 6.00.

Assuming that the weight is normally distributed with mean μ , and variance σ^2 , find a 95% confidence interval for μ .

Solved example 2 (continued)

Solution: Using the data we obtain,

$$\begin{aligned}\bar{X} &= \frac{1}{10}(7.72 + 9.58 + 12.38 + 7.77 + 11.27 + 8.80 + 11.10 + 7.80 + 10.17 + 6.00) \\ &= 9.259\end{aligned}$$

and

$$\begin{aligned}S^2 &= \frac{1}{9}\{(7.72 - 9.259)^2 + (9.58 - 9.259)^2 + (12.38 - 9.259)^2 + (7.77 - 9.259)^2 + (11.27 - 9.259)^2 + (8.80 - 9.259)^2 \\ &\quad + (11.10 - 9.259)^2 + (7.80 - 9.259)^2 + (10.17 - 9.259)^2 + (6.00 - 9.259)^2\} \\ &= 3.96\end{aligned}$$

Solved example 2 (continued)

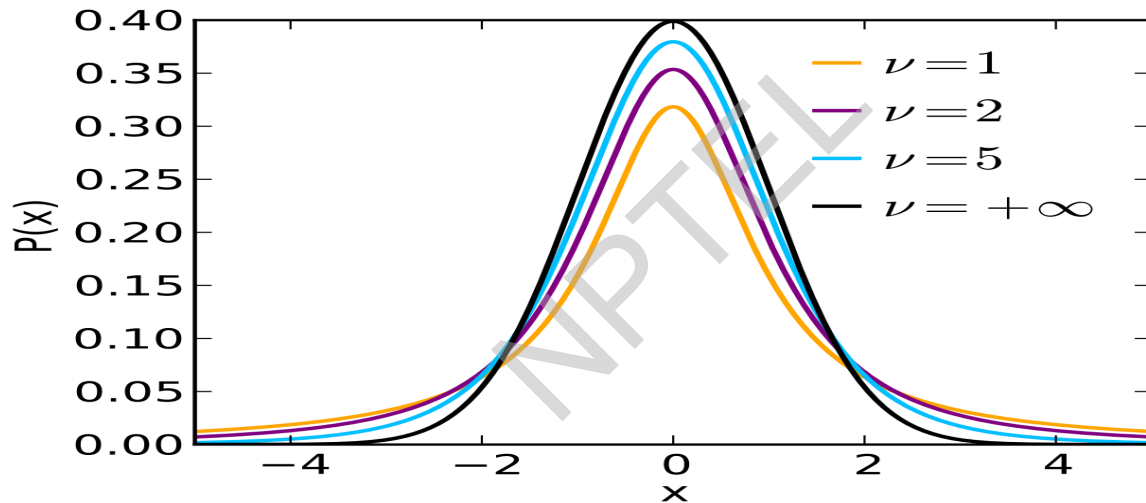
Here $n = 10$, $1 - \alpha = 0.95$; so $\alpha = 0.05$.

Now, from the table $t_{0.025,9} = 2.262$.

Therefore a 95% confidence interval of μ is given by

$$\begin{aligned} &= \left[\bar{X} - t_{\frac{\alpha}{2}, n-1} \frac{S}{\sqrt{n}}, \bar{X} + t_{\frac{\alpha}{2}, n-1} \frac{S}{\sqrt{n}} \right] \\ &= \left[9.26 - 2.26 \times \frac{\sqrt{3.96}}{\sqrt{10}}, 9.26 + 2.26 \times \frac{\sqrt{3.96}}{\sqrt{10}} \right] \\ &= [7.84, 10.68] \end{aligned}$$

Graph of t- distribution



ν denotes the degrees of freedom .

Confidence Interval for Variance

- Let $X_1, \dots, X_n$ be a random sample from the $\mathcal{N}(\mu, \sigma^2)$ distribution.
- We need to estimate the parameter σ^2 and assume that μ is unknown.
- Consider the random variable

$$Q = \frac{(n-1)S^2}{\sigma^2}$$

where $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$.

- Then $Q \sim \chi^2(n-1)$. (Note that $\chi_n^2 \sim \Gamma(\frac{n}{2}, 2)$ as distribution.)

Confidence Interval for Variance (continued)

- For a $100(1 - \alpha)\%$, (where $0 < \alpha < 1$), confidence interval estimation we have,

$$P(\chi_{1-\frac{\alpha}{2}, n-1}^2 \leq Q \leq \chi_{\frac{\alpha}{2}, n-1}^2) = 1 - \alpha$$
$$\text{or, } P\left(\chi_{1-\frac{\alpha}{2}, n-1}^2 \leq \frac{(n-1)S^2}{\sigma^2} \leq \chi_{\frac{\alpha}{2}, n-1}^2\right) = 1 - \alpha$$
$$\text{or, } P\left(\frac{(n-1)S^2}{\chi_{\frac{\alpha}{2}, n-1}^2} \leq \sigma^2 \leq \frac{(n-1)S^2}{\chi_{1-\frac{\alpha}{2}, n-1}^2}\right) = 1 - \alpha$$

- Therefore $\left[\frac{(n-1)S^2}{\chi_{\frac{\alpha}{2}, n-1}^2}, \frac{(n-1)S^2}{\chi_{1-\frac{\alpha}{2}, n-1}^2}\right]$ is $100(1 - \alpha)\%$ confidence interval for σ^2 .

Solved Example 3

Question: Assume the data set in the solved example 1. Find a 95% confidence interval for σ^2 , where μ and σ are unknown.

Solution: Remember that $\bar{X} = 9.259$ and $S^2 = 3.96$.

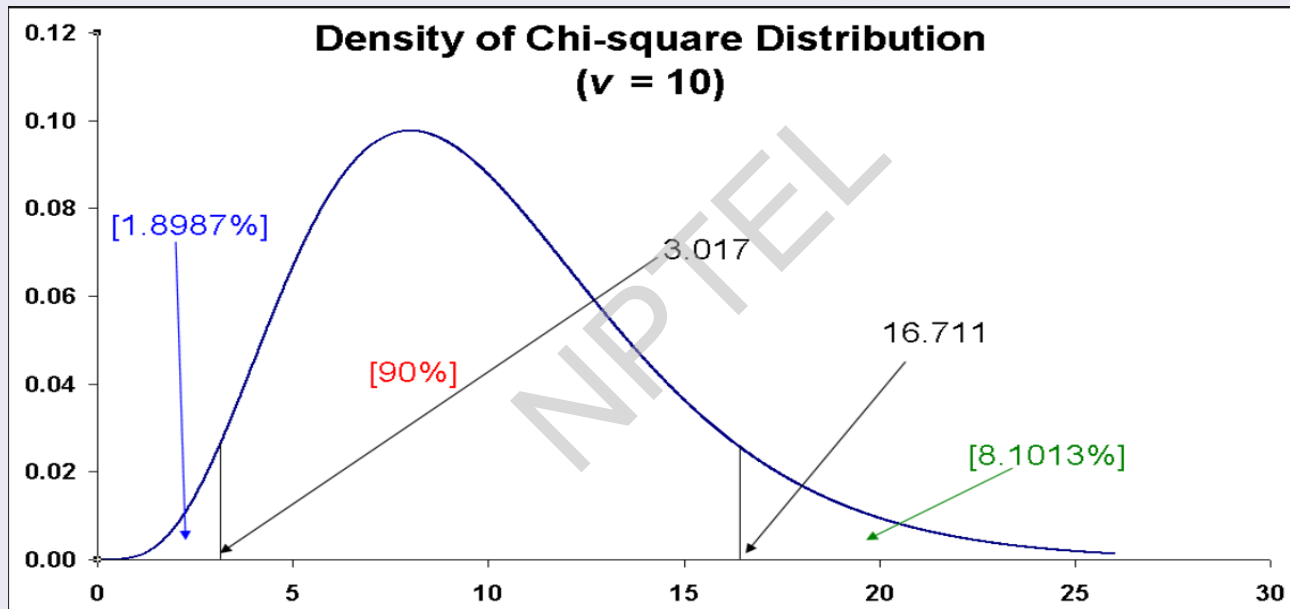
Now $\chi^2_{0.025,9} = 19.02$ and $\chi^2_{0.075,9} = 2.70$.

Solved Example 3 (continued)

Therefore a 95% confidence interval for σ^2 is

$$\begin{aligned} &= \left[\frac{(n-1)S^2}{\chi^2_{\frac{\alpha}{2}, n-1}}, \frac{(n-1)S^2}{\chi^2_{1-\frac{\alpha}{2}, n-1}} \right] \\ &= \left[\frac{9 \times 3.96}{19.02}, \frac{9 \times 3.96}{2.70} \right] \\ &= [1.87, 13.20] \end{aligned}$$

Confidence Interval for the χ^2 distribution



ν denotes the degrees of freedom .

Confidence Interval for the Bernoulli Distribution

- Let $X_1, \dots, X_n$ be samples form a Bernoulli distribution with the probability of success p .
- Recall that the $E(X_i) = p$ and $Var(X_i) = p(1 - p)$ for all i .
- We want to estimate p .

Confidence Interval for the Bernoulli Distribution (continued)

- Let $\hat{p} = \frac{1}{n} \sum_{i=1}^n X_i$.
- Consider the random variable

$$Z = \frac{\hat{p} - p}{\sqrt{p(1-p)/n}}$$

- Then Z follows the standard normal distribution for large n .

Confidence Interval for the Bernoulli Distribution (continued)

- For $100(1 - \alpha)\%$ confidence interval, significance level = $1 - (1 - \alpha) = \alpha$.
For the interval estimation (two tailed) we have,

$$P(-z_{\frac{\alpha}{2}} \leq Z \leq z_{\frac{\alpha}{2}}) = 1 - \alpha$$

$$\text{or, } P\left(-z_{\frac{\alpha}{2}} \leq \frac{\hat{p} - p}{\sqrt{p(1-p)/n}} \leq z_{\frac{\alpha}{2}}\right) = 1 - \alpha$$

$$\text{or, } P\left(-z_{\frac{\alpha}{2}} \sqrt{p(1-p)/n} \leq \hat{p} - p \leq z_{\frac{\alpha}{2}} \sqrt{p(1-p)/n}\right) = 1 - \alpha$$

$$\text{or, } P\left(\hat{p} - z_{\frac{\alpha}{2}} \sqrt{p(1-p)/n} \leq p \leq \hat{p} + z_{\frac{\alpha}{2}} \sqrt{p(1-p)/n}\right) = 1 - \alpha$$

Confidence Interval for the Bernoulli Distribution (continued)

For large enough n , we replace p by $\hat{p}$ in

$$\left[\hat{p} - z_{\frac{\alpha}{2}} \sqrt{p(1-p)/n}, \hat{p} + z_{\frac{\alpha}{2}} \sqrt{p(1-p)/n} \right]$$

we get

$$\left[\hat{p} - z_{\frac{\alpha}{2}} \sqrt{\hat{p}(1-\hat{p})/n}, \hat{p} + z_{\frac{\alpha}{2}} \sqrt{\hat{p}(1-\hat{p})/n} \right]$$

is a $100(1 - \alpha)\%$ confidence interval for p .

References

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Thank you





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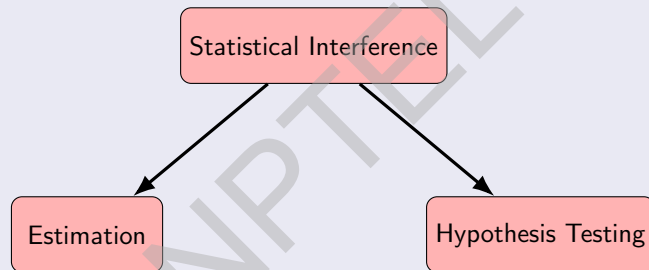
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Module 03: Statistical Inference 2

Lecture 02: Interval Estimation 2



Statistical Inference



Estimation

Tries to predict a value related to the population using a sample statistic $T(X_1, \dots, X_n)$.

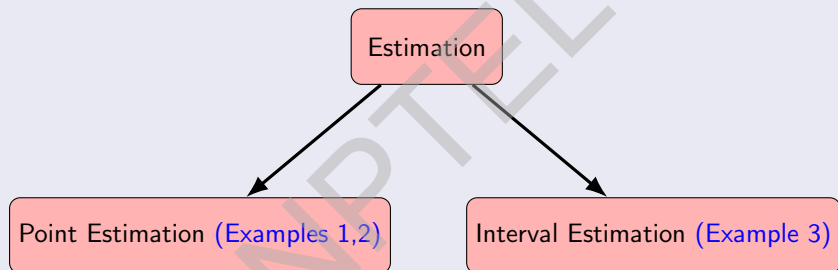
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- Sample Statistic: A function of sample $T(X_1, \dots, X_n)$.

Estimation (continued)

Example:

- 1 Predict the head probability of a coin using the number of heads in n tosses.
- 2 Predict the mean of a Normal Distribution population by $\frac{X_1 + \dots + X_n}{n}$ where $X_1, \dots, X_n$ is a sample from that.
- 3 Try to predict an interval within the variance of a normal distribution that is very likely to belong using $\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$ where $X_1, \dots, X_n$ is the sample and $\bar{X} = \sum_{i=1}^n X_i$, the sample mean.

Estimation (continued)

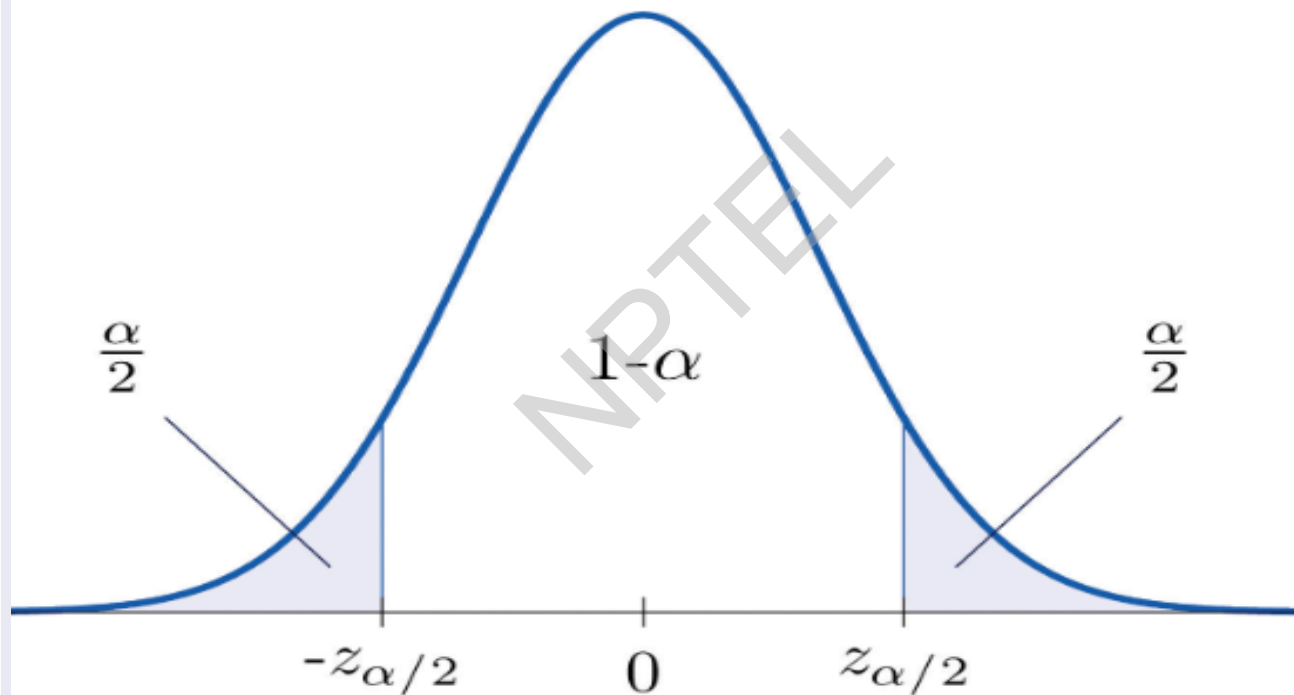


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$$P(\theta \in [c_1(X_1, \dots, X_n), c_2(X_1, \dots, X_n)]) = 1 - \alpha$$

where $c_1(X_1, \dots, X_n) < c_2(X_1, \dots, X_n)$ and $c_1(X_1, \dots, X_n)$ & $c_2(X_1, \dots, X_n)$ are two functions of $X_1, \dots, X_n$.



Useful Results

- **The Weak Law of Large Numbers (WLLN):** Let $X_1, \dots, X_n$ be i.i.d. random variables with a finite expected value μ . Then for any $\epsilon > 0$

$$\lim_{n \rightarrow \infty} P(|\bar{X} - \mu| \geq \epsilon) = 0.$$

where $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$

- Also

$$\begin{aligned} \text{Var}(\bar{X}) &= \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) \\ &= \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) \\ &= \frac{n\sigma^2}{n^2} \\ &= \frac{\sigma^2}{n} \end{aligned}$$

Useful Results

- **Central Limit Theorem (CLT):** Let $X_1, \dots, X_n$ be i.i.d. random variables with expected value $\mu < \infty$ and variance $\sigma^2 < \infty$. Then the random variable

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

is a standard normal variate as $n \rightarrow \infty$.

Useful Results (continued)

- Let $X_1, \dots, X_n$ be i.i.d $\mathcal{N}(\mu, \sigma^2)$ random variables. Also, let S^2 be the sample variance for the random sample. Then the random variable T defined as

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

has t - distribution with $n - 1$ degrees of freedom, i.e., $T \sim t_{n-1}$.

Recall that the pdf of that distribution is

$$\frac{\Gamma(\frac{n+1}{2})}{\sqrt{\pi n} \Gamma(\frac{n}{2})} \left(1 + \frac{x^2}{n}\right)^{-\frac{n+1}{2}}$$

where $n > 0$ is the degrees of freedom and $x \in (-\infty, \infty)$

Useful Results (continued)

- Let $X_1, \dots, X_n$ be i.i.d. $\mathcal{N}(\mu, \sigma^2)$ random variables. Also let S^2 be the sample variance. If

$$Y = \frac{(n-1)S^2}{\sigma^2}$$

has the χ^2 distribution with $(n-1)$ degrees of freedom.

Confidence Interval for Mean When Variance is Known

- Let $X_1, \dots, X_n$ be a random sample from the $\mathcal{N}(\mu, \sigma^2)$ distribution.
- We want to estimate the parameter μ assume that σ^2 is known.
- Construct the random variable

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- Z follows $\mathcal{N}(0, 1)$ distribution.

Confidence Interval for Mean When Variance is Known (continued)

- For a $100(1 - \alpha)\%$ confidence interval, (where $0 < \alpha < 1$), significance level = $1 - (1 - \alpha) = \alpha$.
For the interval estimation (two tailed) we have,

$$P(-z_{\frac{\alpha}{2}} \leq Z \leq z_{\frac{\alpha}{2}}) = 1 - \alpha$$

$$\text{or, } P\left(-z_{\frac{\alpha}{2}} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq z_{\frac{\alpha}{2}}\right) = 1 - \alpha$$

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$$\text{or, } P\left(\bar{X} - z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$

- Therefore $\left[\bar{X} - z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}\right]$ is a $100(1 - \alpha)\%$ confidence interval for μ .

Solved Example 1

Question : Suppose the following five numbers were sampled from a normal distribution with a standard deviation 2.5 : 2,3,5,6 and 9. Compute a 95% confidence interval for mean.

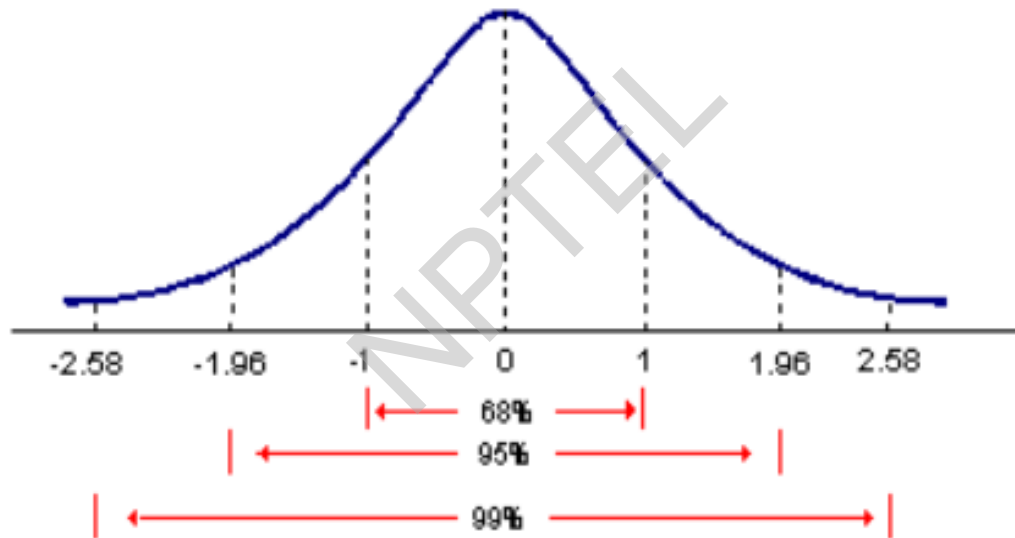
Solution: Here

$$\begin{aligned}\bar{X} &= \frac{2+3+5+6+9}{5} \\ &= 5\end{aligned}$$

and $\sigma = 2.5$. From table $z_{0.025} = 1.96$ Therefore a 95% confidence interval for μ is

$$\begin{aligned}&= \left[\bar{X} - z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \right] \\ &= \left[5 - \frac{1.96 \times 2.5}{\sqrt{5}}, 5 + \frac{1.96 \times 2.5}{\sqrt{5}} \right] \\ &= [2.81, 7.19]\end{aligned}$$

Standard Normal Distribution



Remarks

If $X_1, \dots, X_n$ are random samples from some distribution with mean μ and variance σ^2 . Then for large n , the random variable

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follows $\mathcal{N}(0, 1)$ distribution where $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ and we can estimate the parameter as before.

Confidence Interval for Mean When Variance is Unknown

- Let $X_1, \dots, X_n$ be a random sample from the $\mathcal{N}(\mu, \sigma^2)$ distribution.
- We want to estimate the parameter μ and assume that σ^2 is unknown.
- Construct the random variable

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

where $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ is the sample variance.

- Then T has t - distribution with $n - 1$ degrees of freedom.

Confidence Interval for Mean When Variance is Unknown (continued)

- For a $100(1 - \alpha)\%$, where $(0 < \alpha < 1)$, confidence interval estimation (two tailed) we have,

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$$\text{or, } P\left(-t_{\frac{\alpha}{2}, n-1} \leq \frac{\bar{X} - \mu}{S/\sqrt{n}} \leq t_{\frac{\alpha}{2}, n-1}\right) = 1 - \alpha$$

$$\text{or, } P\left(-t_{\frac{\alpha}{2}, n-1} \frac{S}{\sqrt{n}} \leq \bar{X} - \mu \leq t_{\frac{\alpha}{2}, n-1} \frac{S}{\sqrt{n}}\right) = 1 - \alpha$$

$$\text{or, } P\left(\bar{X} - t_{\frac{\alpha}{2}, n-1} \frac{S}{\sqrt{n}} \leq \mu \leq \bar{X} + t_{\frac{\alpha}{2}, n-1} \frac{S}{\sqrt{n}}\right) = 1 - \alpha$$

- Therefore $\left[\bar{X} - t_{\frac{\alpha}{2}, n-1} \frac{S}{\sqrt{n}}, \bar{X} + t_{\frac{\alpha}{2}, n-1} \frac{S}{\sqrt{n}}\right]$ is a $100(1 - \alpha)\%$ confidence interval for μ .

Solved example 2

Question : A farmer weights 10 randomly chosen watermelons from his farm and he obtains the following values (in lbs):

7.72 9.58 12.38 7.77 11.27 8.80 11.10 7.80 10.17 6.00.

Assuming that the weight is normally distributed with mean μ , and variance σ^2 , find a 95% confidence interval for μ .

Solved example 2 (continued)

Solution: Using the data we obtain,

$$\begin{aligned}\bar{X} &= \frac{1}{10}(7.72 + 9.58 + 12.38 + 7.77 + 11.27 + 8.80 + 11.10 + 7.80 + 10.17 + 6.00) \\ &= 9.259\end{aligned}$$

and

$$\begin{aligned}S^2 &= \frac{1}{9}\{(7.72 - 9.259)^2 + (9.58 - 9.259)^2 + (12.38 - 9.259)^2 + (7.77 - 9.259)^2 + (11.27 - 9.259)^2 + (8.80 - 9.259)^2 \\ &\quad + (11.10 - 9.259)^2 + (7.80 - 9.259)^2 + (10.17 - 9.259)^2 + (6.00 - 9.259)^2\} \\ &= 3.96\end{aligned}$$

Solved example 2 (continued)

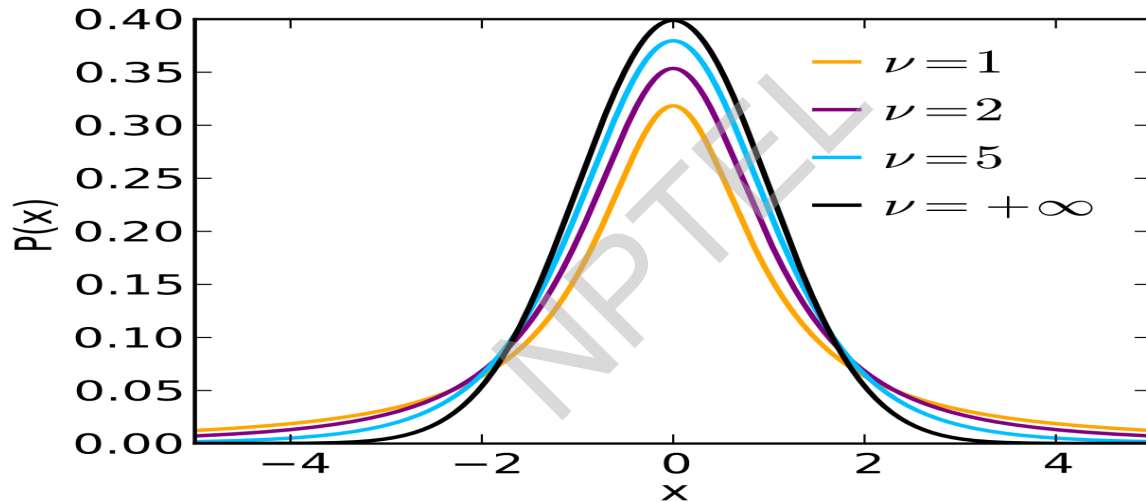
Here $n = 10$, $1 - \alpha = 0.95$; so $\alpha = 0.05$.

Now, from the table $t_{0.025,9} = 2.262$.

Therefore a 95% confidence interval of μ is given by

$$\begin{aligned} &= \left[\bar{X} - t_{\frac{\alpha}{2}, n-1} \frac{S}{\sqrt{n}}, \bar{X} + t_{\frac{\alpha}{2}, n-1} \frac{S}{\sqrt{n}} \right] \\ &= \left[9.26 - 2.26 \times \frac{\sqrt{3.96}}{\sqrt{10}}, 9.26 + 2.26 \times \frac{\sqrt{3.96}}{\sqrt{10}} \right] \\ &= [7.84, 10.68] \end{aligned}$$

Graph of t- distribution



ν denotes the degrees of freedom .

Confidence Interval for Variance

- Let $X_1, \dots, X_n$ be a random sample from the $\mathcal{N}(\mu, \sigma^2)$ distribution.
- We need to estimate the parameter σ^2 and assume that μ is unknown.
- Consider the random variable

$$Q = \frac{(n-1)S^2}{\sigma^2}$$

where $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$.

- Then $Q \sim \chi^2(n-1)$. (Note that $\chi_n^2 \sim \Gamma(\frac{n}{2}, 2)$ as distribution.)

Confidence Interval for Variance (continued)

- For a $100(1 - \alpha)\%$, (where $0 < \alpha < 1$), confidence interval estimation we have,

$$\begin{aligned} P(\chi_{1-\frac{\alpha}{2}, n-1}^2 \leq Q \leq \chi_{\frac{\alpha}{2}, n-1}^2) &= 1 - \alpha \\ \text{or, } P\left(\chi_{1-\frac{\alpha}{2}, n-1}^2 \leq \frac{(n-1)S^2}{\sigma^2} \leq \chi_{\frac{\alpha}{2}, n-1}^2\right) &= 1 - \alpha \\ \text{or, } P\left(\frac{(n-1)S^2}{\chi_{\frac{\alpha}{2}, n-1}^2} \leq \sigma^2 \leq \frac{(n-1)S^2}{\chi_{1-\frac{\alpha}{2}, n-1}^2}\right) &= 1 - \alpha \end{aligned}$$

- Therefore $\left[\frac{(n-1)S^2}{\chi_{\frac{\alpha}{2}, n-1}^2}, \frac{(n-1)S^2}{\chi_{1-\frac{\alpha}{2}, n-1}^2} \right]$ is $100(1 - \alpha)\%$ confidence interval for σ^2 .

Solved Example 3

Question: Assume the data set in the solved example 1. Find a 95% confidence interval for σ^2 , where μ and σ are unknown.

Solution: Remember that $\bar{X} = 9.259$ and $S^2 = 3.96$.

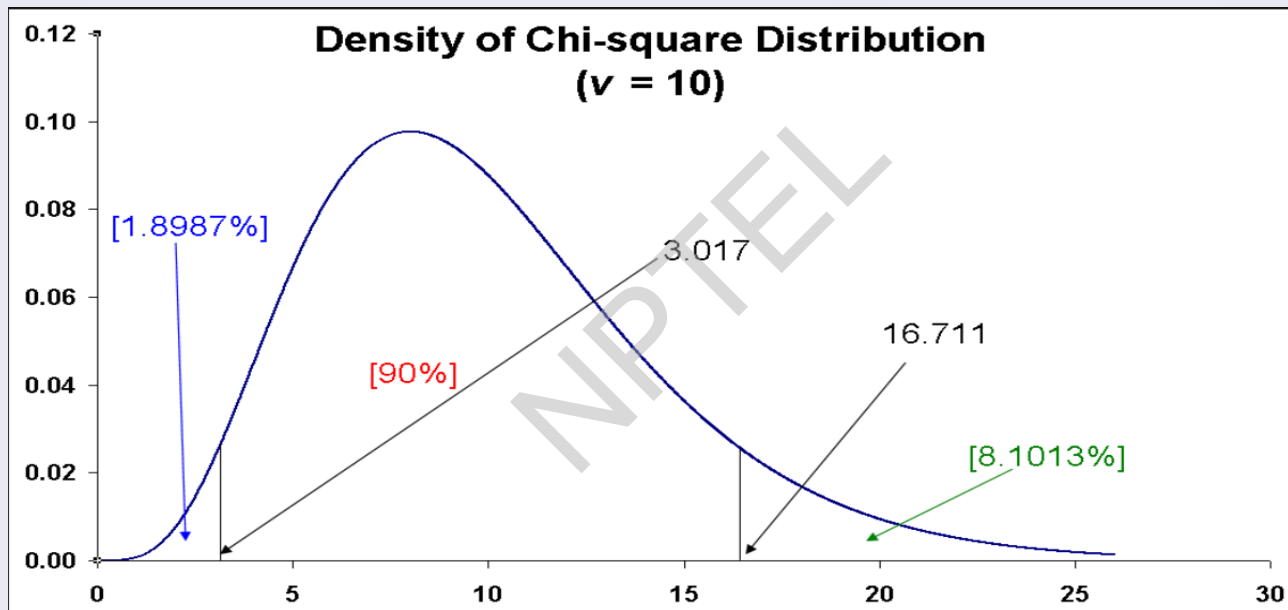
Now $\chi^2_{0.025,9} = 19.02$ and $\chi^2_{0.075,9} = 2.70$.

Solved Example 3 (continued)

Therefore a 95% confidence interval for σ^2 is

$$\begin{aligned} &= \left[\frac{(n-1)S^2}{\chi_{\frac{\alpha}{2}, n-1}^2}, \frac{(n-1)S^2}{\chi_{1-\frac{\alpha}{2}, n-1}^2} \right] \\ &= \left[\frac{9 \times 3.96}{19.02}, \frac{9 \times 3.96}{2.70} \right] \\ &= [1.87, 13.20] \end{aligned}$$

Confidence Interval for the χ^2 distribution



ν denotes the degrees of freedom .

Confidence Interval for the Bernoulli Distribution

- Let $X_1, \dots, X_n$ be samples form a Bernoulli distribution with the probability of success p .
- Recall that the $E(X_i) = p$ and $Var(X_i) = p(1 - p)$ for all i .
- We want to estimate p .

Confidence Interval for the Bernoulli Distribution (continued)

- Let $\hat{p} = \frac{1}{n} \sum_{i=1}^n X_i$.
- Consider the random variable

$$Z = \frac{\hat{p} - p}{\sqrt{p(1-p)/n}}$$

- Then Z follows the standard normal distribution for large n .

Confidence Interval for the Bernoulli Distribution (continued)

- For $100(1 - \alpha)\%$ confidence interval, significance level $= 1 - (1 - \alpha) = \alpha$.
For the interval estimation (two tailed) we have,

$$P(-z_{\frac{\alpha}{2}} \leq Z \leq z_{\frac{\alpha}{2}}) = 1 - \alpha$$

$$\text{or, } P\left(-z_{\frac{\alpha}{2}} \leq \frac{\hat{p} - p}{\sqrt{p(1-p)/n}} \leq z_{\frac{\alpha}{2}}\right) = 1 - \alpha$$

$$\text{or, } P\left(-z_{\frac{\alpha}{2}} \sqrt{p(1-p)/n} \leq \hat{p} - p \leq z_{\frac{\alpha}{2}} \sqrt{p(1-p)/n}\right) = 1 - \alpha$$

$$\text{or, } P\left(\hat{p} - z_{\frac{\alpha}{2}} \sqrt{p(1-p)/n} \leq p \leq \hat{p} + z_{\frac{\alpha}{2}} \sqrt{p(1-p)/n}\right) = 1 - \alpha$$

Confidence Interval for the Bernoulli Distribution (continued)

For large enough n , we replace p by $\hat{p}$ in

$$\left[\hat{p} - z_{\frac{\alpha}{2}} \sqrt{p(1-p)/n}, \hat{p} + z_{\frac{\alpha}{2}} \sqrt{p(1-p)/n} \right]$$

we get

$$\left[\hat{p} - z_{\frac{\alpha}{2}} \sqrt{\hat{p}(1-\hat{p})/n}, \hat{p} + z_{\frac{\alpha}{2}} \sqrt{\hat{p}(1-\hat{p})/n} \right]$$

is a $100(1 - \alpha)\%$ confidence interval for p .

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Thank you





SWAYAM NPTEL COURSE ON STATISTICAL FOUNDATION FOR BIG DATA ANALYSIS

BY
PROF. ARINDAM BANERJEE
Department of Mathematics
INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

Module 03: Statistical Inference 2

Lecture 03: Basics of Hypothesis Testing-I



What is Statistical Inference?

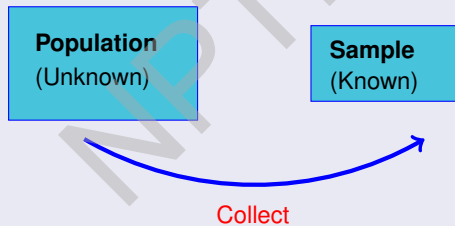
Statistical Inference Process

- Taking decisions about population based on a sample.

Population
(Unknown)

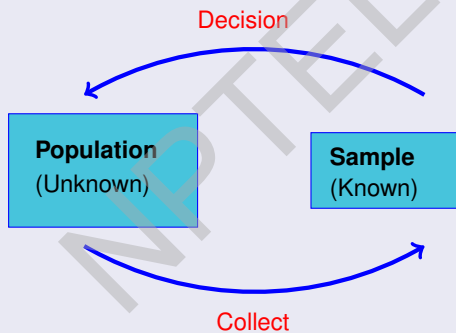
Statistical Inference Process

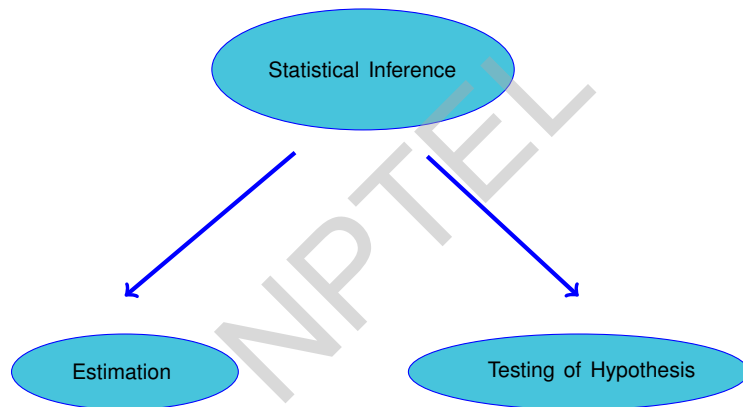
- Taking decisions about population based on a sample.



Statistical Inference Process

- Taking decisions about population based on a sample.





Comparison of Estimation and Hypothesis Testing

Question 1: What is the average height of a person living in Antarctica?	Question 2: Is the average height of a person living in India above 7ft?
Answer 1: No idea	Answer 2: I have some idea but I am not very sure
Approach of study: Estimation	Approach of study: Testing of Hypothesis

Comparison of Estimation and Hypothesis Testing

Question 1: What is the average temperature on the surface of Neptune on January 26, 2026 ?	Question 2: Is the average temperature in Kolkata on July 24, 2026 above 18 degrees Celsius?
Answer 1: No idea	Answer 2: I have some idea but I am not very sure
Approach of study: Estimation	Approach of study: Testing of Hypothesis

Comparison of Estimation and Hypothesis Testing

Question 1: What is the average score of students in a national-level mathematics exam?	Question 2: Do students score more than 95% on average in a national level mathematics exam?
Answer 1: No idea	Answer 2: I have some idea but I am not very sure
Approach of study: Estimation	Approach of study: Testing of Hypothesis

Key Components of a Statistical Test

- Test is a **Rule** where we evaluate an **Assertion** about a **population** against a given **Alternative** statement based on a collected **sample**.
- This assertion and the alternative statement are the **hypotheses** of the test.
- Evaluation is done based on a numerical function of the sample → **Test Statistic**
- Test statistic leads us to take **Decisions**

Type-I and Type-II Error

Decision \ Reality	Assertion is True	Assertion is False
Reject Assertion	Error	No Error
Reject Alternate	No Error	Error

- Two types of Errors can happen
 - Wrongly Rejecting the Assertion
 - wrongly Rejecting the Alternative!!
- More serious error is called Type I Error
- The other one is Type II error ← obviously this depends on the purpose of the test.

Example

Suppose a person with a tumor goes to a doctor. Doctor checks the patient and makes a decision.

Now the doctor can make two kinds of mistakes:

- Fails to diagnose the existing cancer.
- Wrongly diagnose the cancer.

- More Serious Error?
- Failing to diagnose existing cancer ← Type I Error
- Null hypothesis or H_0 : The patient has cancer
- Alternate hypothesis or H_1 : The patient does not have cancer

The job of the hypothesis testing is to decide whether there are overwhelming evidence to reject H_0 !!

Formal Set Up

- **Objective:** To test whether there is an overwhelming evidence to reject a hypothesis H_0 in favour of a stated alternative H_1 based on a sample.
- Chooses a numerical function T of the observables such that one can predict the behaviour of T under H_0 and H_1 .
- Chooses a region C (**Critical region**) such that if the value of $T \in C$, the observations are overwhelmingly more likely to come from H_1 than H_0 .

Type I and Type II Error

- **Type I error:** When H_0 is the true then value of $T \in C$.
- **Type II error:** When H_1 is the true then value of $T \notin C$.

Decision \ Reality	H_0 is True	H_0 is False
	Type I Error	Correct Decision
Reject H_0		
Don't Reject H_0	Correct Decision	Type II Error

Example

Suppose a coin is tossed. The probability of getting head is p .
We want to test: $H_0 : p = 0.5$ and $H_1 = 0.6$.

Rule: H_0 is rejected in favour of H_1 if based on the outcomes of ten tosses of the coin, atleast 7 heads were observed.

Note:

Number of heads $\rightarrow X$
 $X \sim \text{Binomial}(10, p)$.

- $P(\text{Type I Error}) = P(X \geq 7)$ when $p = 0.5$

$$= \left[\binom{10}{7} + \binom{10}{8} + \binom{10}{9} + \binom{10}{10} \right] * (0.5)^{10} \\ = 0.172$$

- $P(\text{Type II Error}) = P(X < 7)$ when $p = 0.6$

$$= 1 - \left[\binom{10}{7} (0.6)^7 \cdot (0.4)^3 + \binom{10}{8} (0.6)^8 \cdot (0.4)^2 + \right. \\ \left. \binom{10}{9} (0.6)^9 \cdot (0.4)^1 + \binom{10}{10} (0.6)^{10} (0.4)^0 \right] \\ = 0.618$$

$H_0 : p = 0.5$ vs $H_1 : p = 0.6$

sample size: 10

Rule $X \geq$	$P(\text{Type I Error})$	$P(\text{Type II Error})$
7	0.171	0.61
6	0.37	0.36
8	0.054	0.83
9	0.01	0.95
10	0.0009	0.99

$$H_0 : p = 0.5 \text{ vs } H_1 : p = 0.6$$

sample size: 100

X	$P(\text{Type I Error})$	$P(\text{Type II Error})$
70	0.000039	0.98
60	0.028	0.46
55	0.18	0.13
56	0.14	0.18
57	0.097	0.24

sample size: 1000

X	$P(\text{Type I Error})$	$P(\text{Type II Error})$
530	0.03	0.0000003
527	0.047	0.0000013

Test for Sample Mean when Population Variance is Known.

Formal Setup

Suppose $X \sim N(\mu, \sigma^2)$ where $\mu \rightarrow$ unknown, $\sigma \rightarrow$ known.
 $X_1, X_2, \dots, X_n$ is an i.i.d. sample from population X which is known.

$$H_0 : \mu = c_1 \quad \text{vs} \quad H_1 : \mu = c_2 \quad \text{with } c_2 > c_1$$

Rule: H_0 is rejected in favour of H_1 if $\bar{X} \geq c_1 + z_\alpha \cdot \frac{\sigma}{\sqrt{n}}$

Test Statistic: $Z = \frac{\bar{X} - c_1}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1)$ under H_0

Note:

σ is known.

$X \sim \mathcal{N}(\mu, \sigma^2)$.

Test for Sample Mean when Population Variance is Known.

Probability of Type I Error

$$\alpha = P(\text{Type I error}) = P(\text{Reject } H_0 \mid \mu = c_1) = P\left(\bar{X} \geq c_1 + z_\alpha \cdot \frac{\sigma}{\sqrt{n}} \mid \mu = c_1\right)$$

Under H_0 , we have: $Z = \frac{\bar{X} - c_1}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1)$. Therefore, $\alpha = P(Z \geq z_\alpha)$.

Note: α is usually chosen by the researcher (e.g. 0.05 or 0.01).

Probability of Type II Error

$$\beta = P(\text{Type II error}) = P(\text{Failing to reject } H_0 \mid \mu = c_2) = P\left(\bar{X} < c_1 + z_\alpha \cdot \frac{\sigma}{\sqrt{n}} \mid \mu = c_2\right)$$

Under H_1 , we have $Z = \frac{\bar{X} - c_2}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1)$

$$\beta = P\left(\bar{X} < c_1 + z_\alpha \cdot \frac{\sigma}{\sqrt{n}} \mid \mu = c_2\right) = P\left(\frac{\bar{X} - c_2}{\sigma/\sqrt{n}} < \frac{c_1 + z_\alpha \cdot \frac{\sigma}{\sqrt{n}} - c_2}{\sigma/\sqrt{n}}\right)$$

Therefore, $\beta = P\left(Z \leq z_\alpha + \frac{c_1 - c_2}{\sigma/\sqrt{n}}\right)$

Example

Suppose we want to test whether a new teaching method has increased student performance. The test scores are normally distributed with a known standard deviation $\sigma = 10$. A random sample of size $n = 25$ is taken.

We want to test:

$$H_0 : \mu = 70 \quad \text{vs} \quad H_1 : \mu = 75$$

Rule: Reject H_0 in favor of H_1 if the sample mean $\bar{X} \geq 73.29$

Z-Test for Mean (Known σ)

Define the test statistic:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{\bar{X} - 70}{2}$$

Rejection rule: Reject H_0 if $\bar{X} \geq 73.29$, which corresponds to:

$$Z \geq \frac{73.29 - 70}{2} = 1.645$$

Error Probabilities

- **Type I Error:** Rejecting H_0 when $\mu = 70$

$$P(\bar{X} \geq 73.29 \mid \mu = 70) = P(Z \geq 1.645) = 0.05$$

- **Type II Error:** Failing to reject H_0 when $\mu = 75$

$$P(\bar{X} < 73.29 \mid \mu = 75) = P\left(Z < \frac{73.29 - 75}{2}\right) = P(Z < -0.855) = 0.196$$

Hypotheses: $H_0 : \mu = 70$ vs $H_1 : \mu = 75$

Sample size: $n = 25$, $\sigma = 10$

Rule $\bar{X} \geq$	Type I Error	Type II Error
73.29	0.05	0.196
74.00	0.0062	0.370
72.00	0.1587	0.0918
71.00	0.3085	0.0446
70.00	0.5	0.0062

Effect of Sample Size with Various Critical Values

Hypotheses: $H_0 : \mu = 70$ vs $H_1 : \mu = 75$

Population standard deviation: $\sigma = 10$

Sample Size n	Critical $\bar{X}$	Type I Error	Type II Error
25	73.29	0.050	0.196
25	74.00	0.0062	0.370
25	72.00	0.1587	0.0918
100	71.65	0.050	0.00003
100	72.00	0.0228	0.0004
100	73.00	0.0013	0.0082

Critical value is calculated using $\bar{X}_{\text{crit}} = \mu_0 + z_{\alpha} \cdot \frac{\sigma}{\sqrt{n}}$, where $z_{0.05} = 1.645$

Test for Sample Mean when Population Variance is Unknown.

Formal Setup

Suppose $X \sim N(\mu, \sigma^2)$ where $\mu \rightarrow$ unknown, $\sigma \rightarrow$ unknown.
 $X_1, X_2, \dots, X_n$ is an i.i.d. sample from population X which is known.

$$H_0 : \mu = c_1 \quad \text{vs} \quad H_1 : \mu = c_2 \quad \text{with } c_2 > c_1$$

Rule: H_0 is rejected in favour of H_1 if $\bar{X} \geq c_1 + t_{\alpha, n-1} \cdot \frac{s}{\sqrt{n}}$

Test Statistic: $T = \frac{\bar{X} - c_1}{s/\sqrt{n}} \sim t_{n-1}$ under H_0

Note:

σ is known.

$X \sim \mathcal{N}(\mu, \sigma^2)$.

Probability of Type I Error

$$\alpha = P(\text{Type I error}) = P(\text{Reject } H_0 \mid \mu = c_1) = P\left(\bar{X} \geq c_1 + t_{\alpha, n-1} \cdot \frac{S}{\sqrt{n}} \mid \mu = c_1\right)$$

Under H_0 , we have: $T = \frac{\bar{X} - c_1}{S/\sqrt{n}} \sim t_{n-1}$. Therefore, $\alpha = P(T \geq t_{\alpha, n-1})$.

Note: α is usually chosen by the researcher (e.g. 0.05 or 0.01).

Probability of Type II Error

$$\beta = P(\text{Type II error}) = P(\text{Failing to reject } H_0 \mid \mu = c_2) = P\left(\bar{X} < c_1 + t_{\alpha, n-1} \cdot \frac{S}{\sqrt{n}} \mid \mu = c_2\right)$$

Under H_1 , we have $T = \frac{\bar{X} - c_2}{S/\sqrt{n}} \sim t_{n-1}$

$$\beta = P\left(\bar{X} < c_1 + t_{\alpha, n-1} \cdot \frac{S}{\sqrt{n}} \mid \mu = c_2\right) = P\left(\frac{\bar{X} - c_2}{S/\sqrt{n}} < \frac{c_1 + t_{\alpha, n-1} \cdot \frac{S}{\sqrt{n}} - c_2}{S/\sqrt{n}}\right)$$

Therefore,
$$\beta = P\left(T \leq t_{\alpha, n-1} + \frac{c_1 - c_2}{S/\sqrt{n}}\right)$$

Example

Suppose a pharmaceutical company is testing whether a new drug reduces systolic blood pressure more than the current standard.

The systolic blood pressure reductions (in mmHg) are assumed to be normally distributed, the sample standard deviation is known to be 5mmHg but the population standard deviation is unknown.

A sample of $n = 25$ patients is taken.

We want to test:

$$H_0 : \mu = 10 \quad \text{vs} \quad H_1 : \mu = 15$$

Rule: Reject H_0 in favor of H_1 if the sample mean $\bar{X} \geq 13.29$

t-Test for Mean (Unknown σ)

Since the population standard deviation is unknown, we use:

$$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}$$

For $n = 25$, degrees of freedom = 24. At $\alpha = 0.05$, one-sided test:

$$t_{\alpha, n-1} = 1.711 \Rightarrow \text{Reject } H_0 \text{ if } t \geq 1.711$$

Assuming sample standard deviation $s = 5$, then:

$$\bar{X}_{\text{crit}} = 10 + 1.711 \cdot \frac{5}{\sqrt{25}} = 10 + 1.711 = 11.711$$

Error Probabilities (Assuming $s = 5$)

- Type I Error:

$$P(t \geq 1.711 \mid \mu = 10) = 0.05$$

- Type II Error:

$$P(t < 1.711 \mid \mu = 15)$$

When $\mu = 15$, mean shift:

$$t = \frac{11.711 - 15}{5/\sqrt{25}} = \frac{-3.289}{1} = -3.289 \Rightarrow P(t < -3.289) \approx 0.0006$$

Hypotheses: $H_0 : \mu = 10$ vs $H_1 : \mu = 15$

Sample size: $n = 25, s = 5$, Degrees of freedom = 24

Rule $\bar{X} \geq$	Type I Error	Type II Error
11.71	0.050	0.0006
12.00	0.025	0.0019
13.00	0.002	0.056
14.00	0.00007	0.308
15.00	0.000003	0.5

Effect of Sample Size with Various Critical Values

Hypotheses: $H_0 : \mu = 10$ vs $H_1 : \mu = 15$

Assume $s = 5$, Use appropriate t_{crit} for each n

Sample Size n	Critical $\bar{X}$	Type I Error	Type II Error
25	11.71	0.050	0.0006
50	11.40	0.050	0.0000005
75	11.29	0.050	$< 10^{-9}$
100	11.23	0.050	$< 10^{-12}$

Critical value: $\bar{X}_{\text{crit}} = \mu_0 + t_{\alpha} \cdot \frac{s}{\sqrt{n}}$

Test for Sample Mean when Population Variance is Known.

Formal Setup

Suppose $X \sim N(\mu, \sigma^2)$ where $\mu \rightarrow$ unknown, $\sigma \rightarrow$ known.
 $X_1, X_2, \dots, X_n$ is an i.i.d. sample from population X which is known.
We want to test: $H_0 : \mu = c$ versus $H_1 : \mu > c$.

Question

If the null hypothesis is true, how likely is it that we would have seen a sample mean as large as $\bar{X}$?

Test for Sample Mean when Population Variance is Known.

Answer

Assume that empirical values of $X_1, X_2, \dots, X_n$ are known.
Let $Y_1, Y_2, \dots, Y_n$ be any i.i.d. sample from population X .

Note:

$$X \sim N(\mu, \sigma^2)$$

$\bar{Y} = \frac{Y_1 + Y_2 + \dots + Y_n}{n}$ is a random variable.

$\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n}$ is the deterministic observed sample average.

Test for Sample Mean when Population Variance is Known.

Answer (Continued)

If the null hypothesis is true, $Z = \frac{\sqrt{n}(\bar{Y} - c)}{\sigma} \sim N(0, 1)$.

$$P(\bar{Y} \geq \bar{X}) = P\left(\frac{\sqrt{n}(\bar{Y} - c)}{\sigma} \geq \frac{\sqrt{n}(\bar{X} - c)}{\sigma}\right) = P\left(Z \geq \frac{\sqrt{n}(\bar{X} - c)}{\sigma}\right)$$

Note:

In practice, before a sample is taken, a *level of significance* $\alpha \in (0, 1)$ is typically fixed.

Definition of p-value

General Idea

The **p-value** is the probability, assuming the null hypothesis H_0 is true, of observing a test statistic at least as extreme as the one calculated from the sample.

In Our Current Example

$\bar{X}$ is the observed sample mean.

$$\text{p-value} = P(\bar{Y} \geq \bar{X} | H_0) = P\left(Z \geq \frac{\sqrt{n}(\bar{X} - c)}{\sigma}\right)$$

Decision:

- If $P(\bar{Y} \geq \bar{X}) < \alpha$ then the sample average is far away from assumed mean c .
 $H_0 : \mu = c$ is judged to be incorrect. **We reject the null hypothesis.**
- If $P(\bar{Y} \geq \bar{X}) \geq \alpha$ then the sample average seems to be consistent with the assumption $\mu = c$. **We do not reject the null hypothesis.**

Example

- Toss a coin 10 times.
- Test: $H_0 : p = 0.5$ vs. $H_1 : p = 0.6$
- **Rule:** Reject H_0 if ≥ 7 heads observed.

p-value = $P(X \geq 7 \mid p = 0.5)$ where $X \sim \text{Binomial}(10, 0.5)$

$$P(X \geq 7 \mid p = 0.5) = \sum_{k=7}^{10} \binom{10}{k} (0.5)^{10} = \frac{11}{64} \approx \boxed{0.1719}$$

References

- 1 Fundamentals of Mathematical Statistics (tenth revised edition), by S.C. Gupta and V.K. Kapoor, Sultan Chand & Sons.
- 2 Probability and Statistics with Examples using R, by Siva Athreya, Deepayan Sarkar, and Steve Tanner, <https://psweur.github.io> (2024).

Thank you





SWAYAM NPTEL COURSE ON STATISTICAL FOUNDATION FOR BIG DATA ANALYSIS

BY
PROF. ARINDAM BANERJEE
Department of Mathematics
INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

Module 03: Statistical Inference 2

Lecture 04: Basics of Hypothesis Testing-II



What is Statistical Inference?

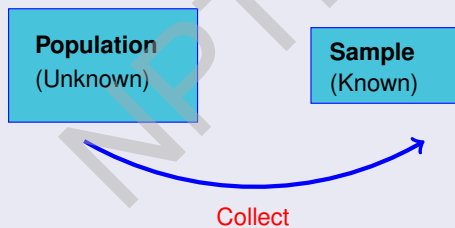
Statistical Inference Process

- Taking decisions about population based on a sample.

Population
(Unknown)

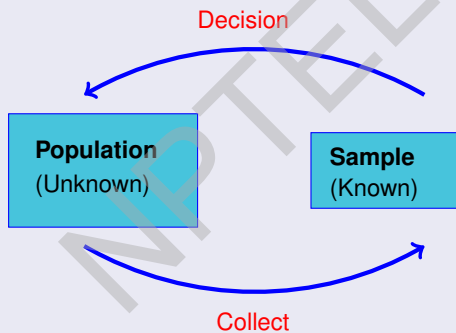
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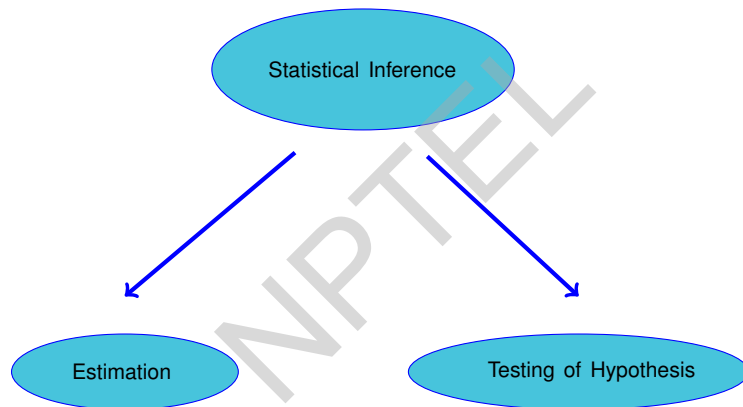
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Statistical Inference Process

- Taking decisions about population based on a sample.





Comparison of Estimation and Hypothesis Testing

Question 1: What is the average height of a person living in Antarctica?	Question 2: Is the average height of a person living in India above 7ft?
Answer 1: No idea	Answer 2: I have some idea but I am not very sure
Approach of study: Estimation	Approach of study: Testing of Hypothesis

Comparison of Estimation and Hypothesis Testing

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Comparison of Estimation and Hypothesis Testing

Question 1: What is the average score of students in a national-level mathematics exam?	Question 2: Do students score more than 95% on average in a national level mathematics exam?
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Key Components of a Statistical Test

- Test is a **Rule** where we evaluate an **Assertion** about a **population** against a given **Alternative** statement based on a collected **sample**.
- This assertion and the alternative statement are the **hypotheses** of the test.
- Evaluation is done based on a numerical function of the sample → **Test Statistic**
- Test statistic leads us to take **Decisions**

Type-I and Type-II Error

Decision \ Reality	Assertion is True	Assertion is False
Reject Assertion	Error	No Error
Reject Alternate	No Error	Error

- Two types of Errors can happen
 - Wrongly Rejecting the Assertion
 - wrongly Rejecting the Alternative!!
- More serious error is called Type I Error
- The other one is Type II error ← obviously this depends on the purpose of the test.

Example

Suppose a person with a tumor goes to a doctor. Doctor checks the patient and makes a decision.

Now the doctor can make two kinds of mistakes:

- Fails to diagnose the existing cancer.
- Wrongly diagnose the cancer.

- More Serious Error?
- Failing to diagnose existing cancer ← Type I Error
- Null hypothesis or H_0 : The patient has cancer
- Alternate hypothesis or H_1 : The patient does not have cancer

The job of the hypothesis testing is to decide whether there are overwhelming evidence to reject H_0 !!

Formal Set Up

- **Objective:** To test whether there is an overwhelming evidence to reject a hypothesis H_0 in favour of a stated alternative H_1 based on a sample.
- Chooses a numerical function T of the observables such that one can predict the behaviour of T under H_0 and H_1 .
- Chooses a region C (**Critical region**) such that if the value of $T \in C$, the observations are overwhelmingly more likely to come from H_1 than H_0 .

Type I and Type II Error

- **Type I error:** When H_0 is the true then value of $T \in C$.
- **Type II error:** When H_1 is the true then value of $T \notin C$.

Decision \ Reality	H_0 is True	H_0 is False
	Type I Error	Correct Decision
Reject H_0	Type I Error	Correct Decision
Don't Reject H_0	Correct Decision	Type II Error

Example

Suppose a coin is tossed. The probability of getting head is p .
We want to test: $H_0 : p = 0.5$ and $H_1 = 0.6$.

Rule: H_0 is rejected in favour of H_1 if based on the outcomes of ten tosses of the coin, atleast 7 heads were observed.

Note:

Number of heads $\rightarrow X$
 $X \sim \text{Binomial}(10, p)$.

- $P(\text{Type I Error}) = P(X \geq 7)$ when $p = 0.5$

$$= \left[\binom{10}{7} + \binom{10}{8} + \binom{10}{9} + \binom{10}{10} \right] * (0.5)^{10} \\ = 0.172$$

- $P(\text{Type II Error}) = P(X < 7)$ when $p = 0.6$

$$= 1 - \left[\binom{10}{7} (0.6)^7 \cdot (0.4)^3 + \binom{10}{8} (0.6)^8 \cdot (0.4)^2 + \right. \\ \left. \binom{10}{9} (0.6)^9 \cdot (0.4)^1 + \binom{10}{10} (0.6)^{10} (0.4)^0 \right] \\ = 0.618$$

$H_0 : p = 0.5$ vs $H_1 : p = 0.6$

sample size: 10

Rule $X \geq$	$P(\text{Type I Error})$	$P(\text{Type II Error})$
7	0.171	0.61
6	0.37	0.36
8	0.054	0.83
9	0.01	0.95
10	0.0009	0.99

$$H_0 : p = 0.5 \text{ vs } H_1 : p = 0.6$$

sample size: 100

X	$P(\text{Type I Error})$	$P(\text{Type II Error})$
70	0.000039	0.98
60	0.028	0.46
55	0.18	0.13
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57	0.097	0.24

sample size: 1000

X	$P(\text{Type I Error})$	$P(\text{Type II Error})$
530	0.03	0.0000003
527	0.047	0.0000013

Test for Sample Mean when Population Variance is Known.

Formal Setup

Suppose $X \sim N(\mu, \sigma^2)$ where $\mu \rightarrow$ unknown, $\sigma \rightarrow$ known.
 $X_1, X_2, \dots, X_n$ is an i.i.d. sample from population X which is known.

$$H_0 : \mu = c_1 \quad \text{vs} \quad H_1 : \mu = c_2 \quad \text{with } c_2 > c_1$$

Rule: H_0 is rejected in favour of H_1 if $\bar{X} \geq c_1 + z_\alpha \cdot \frac{\sigma}{\sqrt{n}}$

Test Statistic: $Z = \frac{\bar{X} - c_1}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1)$ under H_0

Note:

σ is known.

$X \sim \mathcal{N}(\mu, \sigma^2)$.

Test for Sample Mean when Population Variance is Known.

Probability of Type I Error

$$\alpha = P(\text{Type I error}) = P(\text{Reject } H_0 \mid \mu = c_1) = P\left(\bar{X} \geq c_1 + z_\alpha \cdot \frac{\sigma}{\sqrt{n}} \mid \mu = c_1\right)$$

Under H_0 , we have: $Z = \frac{\bar{X} - c_1}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1)$. Therefore, $\alpha = P(Z \geq z_\alpha)$.

Note: α is usually chosen by the researcher (e.g. 0.05 or 0.01).

Probability of Type II Error

$$\beta = P(\text{Type II error}) = P(\text{Failing to reject } H_0 \mid \mu = c_2) = P\left(\bar{X} < c_1 + z_\alpha \cdot \frac{\sigma}{\sqrt{n}} \mid \mu = c_2\right)$$

Under H_1 , we have $Z = \frac{\bar{X} - c_2}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1)$

$$\beta = P\left(\bar{X} < c_1 + z_\alpha \cdot \frac{\sigma}{\sqrt{n}} \mid \mu = c_2\right) = P\left(\frac{\bar{X} - c_2}{\sigma/\sqrt{n}} < \frac{c_1 + z_\alpha \cdot \frac{\sigma}{\sqrt{n}} - c_2}{\sigma/\sqrt{n}}\right)$$

Therefore,
$$\beta = P\left(Z \leq z_\alpha + \frac{c_1 - c_2}{\sigma/\sqrt{n}}\right)$$

Example

Suppose we want to test whether a new teaching method has increased student performance. The test scores are normally distributed with a known standard deviation $\sigma = 10$. A random sample of size $n = 25$ is taken.

We want to test:

$$H_0 : \mu = 70 \quad \text{vs} \quad H_1 : \mu = 75$$

Rule: Reject H_0 in favor of H_1 if the sample mean $\bar{X} \geq 73.29$

Z-Test for Mean (Known σ)

Define the test statistic:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{\bar{X} - 70}{2}$$

Rejection rule: Reject H_0 if $\bar{X} \geq 73.29$, which corresponds to:

$$Z \geq \frac{73.29 - 70}{2} = 1.645$$

Error Probabilities

- **Type I Error:** Rejecting H_0 when $\mu = 70$

$$P(\bar{X} \geq 73.29 \mid \mu = 70) = P(Z \geq 1.645) = 0.05$$

- **Type II Error:** Failing to reject H_0 when $\mu = 75$

$$P(\bar{X} < 73.29 \mid \mu = 75) = P\left(Z < \frac{73.29 - 75}{2}\right) = P(Z < -0.855) = 0.196$$

Hypotheses: $H_0 : \mu = 70$ vs $H_1 : \mu = 75$

Sample size: $n = 25$, $\sigma = 10$

Rule $\bar{X} \geq$	Type I Error	Type II Error
73.29	0.05	0.196
74.00	0.0062	0.370
72.00	0.1587	0.0918
71.00	0.3085	0.0446
70.00	0.5	0.0062

Effect of Sample Size with Various Critical Values

Hypotheses: $H_0 : \mu = 70$ vs $H_1 : \mu = 75$

Population standard deviation: $\sigma = 10$

Sample Size n	Critical $\bar{X}$	Type I Error	Type II Error
25	73.29	0.050	0.196
25	74.00	0.0062	0.370
25	72.00	0.1587	0.0918
100	71.65	0.050	0.00003
100	72.00	0.0228	0.0004
100	73.00	0.0013	0.0082

Critical value is calculated using $\bar{X}_{\text{crit}} = \mu_0 + z_{\alpha} \cdot \frac{\sigma}{\sqrt{n}}$, where $z_{0.05} = 1.645$

Test for Sample Mean when Population Variance is Unknown.

Formal Setup

Suppose $X \sim N(\mu, \sigma^2)$ where $\mu \rightarrow$ unknown, $\sigma \rightarrow$ unknown.
 $X_1, X_2, \dots, X_n$ is an i.i.d. sample from population X which is known.

$$H_0 : \mu = c_1 \quad \text{vs} \quad H_1 : \mu = c_2 \quad \text{with } c_2 > c_1$$

Rule: H_0 is rejected in favour of H_1 if $\bar{X} \geq c_1 + t_{\alpha, n-1} \cdot \frac{s}{\sqrt{n}}$

Test Statistic: $T = \frac{\bar{X} - c_1}{s/\sqrt{n}} \sim t_{n-1}$ under H_0

Note:

σ is known.

$X \sim \mathcal{N}(\mu, \sigma^2)$.

Probability of Type I Error

$$\alpha = P(\text{Type I error}) = P(\text{Reject } H_0 \mid \mu = c_1) = P\left(\bar{X} \geq c_1 + t_{\alpha, n-1} \cdot \frac{S}{\sqrt{n}} \mid \mu = c_1\right)$$

Under H_0 , we have: $T = \frac{\bar{X} - c_1}{S/\sqrt{n}} \sim t_{n-1}$. Therefore, $\alpha = P(T \geq t_{\alpha, n-1})$.

Note: α is usually chosen by the researcher (e.g. 0.05 or 0.01).

Probability of Type II Error

$$\beta = P(\text{Type II error}) = P(\text{Failing to reject } H_0 \mid \mu = c_2) = P\left(\bar{X} < c_1 + t_{\alpha, n-1} \cdot \frac{S}{\sqrt{n}} \mid \mu = c_2\right)$$

Under H_1 , we have $T = \frac{\bar{X} - c_2}{S/\sqrt{n}} \sim t_{n-1}$

$$\beta = P\left(\bar{X} < c_1 + t_{\alpha, n-1} \cdot \frac{S}{\sqrt{n}} \mid \mu = c_2\right) = P\left(\frac{\bar{X} - c_2}{S/\sqrt{n}} < \frac{c_1 + t_{\alpha, n-1} \cdot \frac{S}{\sqrt{n}} - c_2}{S/\sqrt{n}}\right)$$

Therefore,
$$\beta = P\left(T \leq t_{\alpha, n-1} + \frac{c_1 - c_2}{S/\sqrt{n}}\right)$$

Example

Suppose a pharmaceutical company is testing whether a new drug reduces systolic blood pressure more than the current standard.

The systolic blood pressure reductions (in mmHg) are assumed to be normally distributed, the sample standard deviation is known to be 5mmHg but the population standard deviation is unknown.

A sample of $n = 25$ patients is taken.

We want to test:

$$H_0 : \mu = 10 \quad \text{vs} \quad H_1 : \mu = 15$$

Rule: Reject H_0 in favor of H_1 if the sample mean $\bar{X} \geq 13.29$

t-Test for Mean (Unknown σ)

Since the population standard deviation is unknown, we use:

$$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}$$

For $n = 25$, degrees of freedom = 24. At $\alpha = 0.05$, one-sided test:

$$t_{\alpha, n-1} = 1.711 \Rightarrow \text{Reject } H_0 \text{ if } t \geq 1.711$$

Assuming sample standard deviation $s = 5$, then:

$$\bar{X}_{\text{crit}} = 10 + 1.711 \cdot \frac{5}{\sqrt{25}} = 10 + 1.711 = 11.711$$

Error Probabilities (Assuming $s = 5$)

- Type I Error:

$$P(t \geq 1.711 \mid \mu = 10) = 0.05$$

- Type II Error:

$$P(t < 1.711 \mid \mu = 15)$$

When $\mu = 15$, mean shift:

$$t = \frac{11.711 - 15}{5/\sqrt{25}} = \frac{-3.289}{1} = -3.289 \Rightarrow P(t < -3.289) \approx 0.0006$$

Hypotheses: $H_0 : \mu = 10$ vs $H_1 : \mu = 15$

Sample size: $n = 25, s = 5$, Degrees of freedom = 24

Rule $\bar{X} \geq$	Type I Error	Type II Error
11.71	0.050	0.0006
12.00	0.025	0.0019
13.00	0.002	0.056
14.00	0.00007	0.308
15.00	0.000003	0.5

Effect of Sample Size with Various Critical Values

Hypotheses: $H_0 : \mu = 10$ vs $H_1 : \mu = 15$

Assume $s = 5$, Use appropriate t_{crit} for each n

Sample Size n	Critical $\bar{X}$	Type I Error	Type II Error
25	11.71	0.050	0.0006
50	11.40	0.050	0.0000005
75	11.29	0.050	$< 10^{-9}$
100	11.23	0.050	$< 10^{-12}$

Critical value: $\bar{X}_{\text{crit}} = \mu_0 + t_{\alpha} \cdot \frac{s}{\sqrt{n}}$

Test for Sample Mean when Population Variance is Known.

Formal Setup

Suppose $X \sim N(\mu, \sigma^2)$ where $\mu \rightarrow$ unknown, $\sigma \rightarrow$ known.
 $X_1, X_2, \dots, X_n$ is an i.i.d. sample from population X which is known.
We want to test: $H_0 : \mu = c$ versus $H_1 : \mu > c$.

Question

If the null hypothesis is true, how likely is it that we would have seen a sample mean as large as $\bar{X}$?

Test for Sample Mean when Population Variance is Known.

Answer

Assume that empirical values of $X_1, X_2, \dots, X_n$ are known.
Let $Y_1, Y_2, \dots, Y_n$ be any i.i.d. sample from population X .

Note:

$$X \sim N(\mu, \sigma^2)$$

$\bar{Y} = \frac{Y_1 + Y_2 + \dots + Y_n}{n}$ is a random variable.

$\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n}$ is the deterministic observed sample average.

Test for Sample Mean when Population Variance is Known.

Answer (Continued)

If the null hypothesis is true, $Z = \frac{\sqrt{n}(\bar{Y} - c)}{\sigma} \sim N(0, 1)$.

$$P(\bar{Y} \geq \bar{X}) = P\left(\frac{\sqrt{n}(\bar{Y} - c)}{\sigma} \geq \frac{\sqrt{n}(\bar{X} - c)}{\sigma}\right) = P\left(Z \geq \frac{\sqrt{n}(\bar{X} - c)}{\sigma}\right)$$

Note:

In practice, before a sample is taken, a *level of significance* $\alpha \in (0, 1)$ is typically fixed.

Definition of p-value

General Idea

The **p-value** is the probability, assuming the null hypothesis H_0 is true, of observing a test statistic at least as extreme as the one calculated from the sample.

In Our Current Example

$\bar{X}$ is the observed sample mean.

$$\text{p-value} = P(\bar{Y} \geq \bar{X} | H_0) = P\left(Z \geq \frac{\sqrt{n}(\bar{X} - c)}{\sigma}\right)$$

Decision:

- If $P(\bar{Y} \geq \bar{X}) < \alpha$ then the sample average is far away from assumed mean c .
 $H_0 : \mu = c$ is judged to be incorrect. **We reject the null hypothesis.**
- If $P(\bar{Y} \geq \bar{X}) \geq \alpha$ then the sample average seems to be consistent with the assumption $\mu = c$. **We do not reject the null hypothesis.**

Example

- Toss a coin 10 times.
- Test: $H_0 : p = 0.5$ vs. $H_1 : p = 0.6$
- **Rule:** Reject H_0 if ≥ 7 heads observed.

p-value = $P(X \geq 7 \mid p = 0.5)$ where $X \sim \text{Binomial}(10, 0.5)$

$$P(X \geq 7 \mid p = 0.5) = \sum_{k=7}^{10} \binom{10}{k} (0.5)^{10} = \frac{11}{64} \approx \boxed{0.1719}$$

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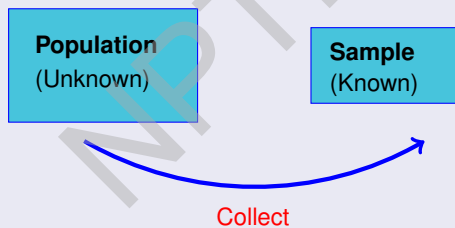
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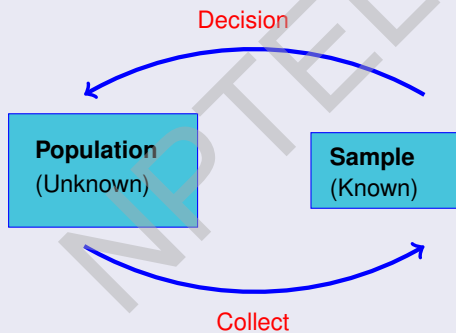
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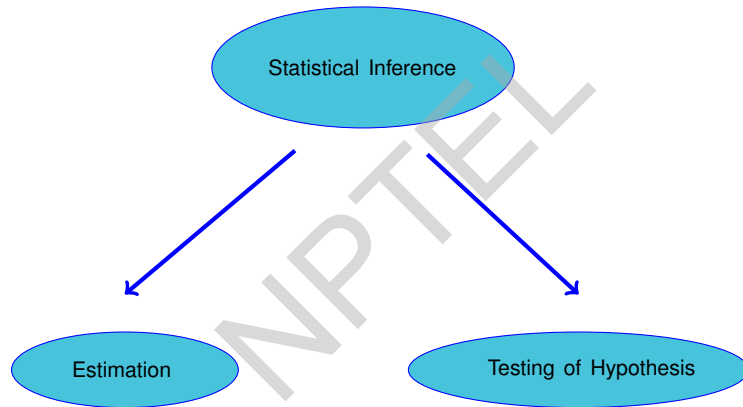
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Suppose a person with a tumor goes to a doctor. Doctor checks the patient and makes a decision.

Now the doctor can make two kinds of mistakes:

- Fails to diagnose the existing cancer.
- Wrongly diagnose the cancer.

- More Serious Error?
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- **Type II error:** When H_1 is the true then value of $T \notin C$.

Decision \ Reality	H_0 is True	H_0 is False
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Don't Reject H_0	Correct Decision	Type II Error

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$$= 1 - \left[\binom{10}{7} (0.6)^7 \cdot (0.4)^3 + \binom{10}{8} (0.6)^8 \cdot (0.4)^2 + \right. \\ \left. \binom{10}{9} (0.6)^9 \cdot (0.4)^1 + \binom{10}{10} (0.6)^{10} (0.4)^0 \right] \\ = 0.618$$

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sample size: 10

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Rule: H_0 is rejected in favour of H_1 if $\bar{X} \geq c_1 + z_\alpha \cdot \frac{\sigma}{\sqrt{n}}$

Test Statistic: $Z = \frac{\bar{X} - c_1}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1)$ under H_0

Note:

σ is known.

$X \sim \mathcal{N}(\mu, \sigma^2)$.

Test for Sample Mean when Population Variance is Known.

Probability of Type I Error

$$\alpha = P(\text{Type I error}) = P(\text{Reject } H_0 \mid \mu = c_1) = P\left(\bar{X} \geq c_1 + z_\alpha \cdot \frac{\sigma}{\sqrt{n}} \mid \mu = c_1\right)$$

Under H_0 , we have: $Z = \frac{\bar{X} - c_1}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1)$. Therefore, $\alpha = P(Z \geq z_\alpha)$.

Note: α is usually chosen by the researcher (e.g. 0.05 or 0.01).

Probability of Type II Error

$$\beta = P(\text{Type II error}) = P(\text{Failing to reject } H_0 \mid \mu = c_2) = P\left(\bar{X} < c_1 + z_\alpha \cdot \frac{\sigma}{\sqrt{n}} \mid \mu = c_2\right)$$

Under H_1 , we have $Z = \frac{\bar{X} - c_2}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1)$

$$\beta = P\left(\bar{X} < c_1 + z_\alpha \cdot \frac{\sigma}{\sqrt{n}} \mid \mu = c_2\right) = P\left(\frac{\bar{X} - c_2}{\sigma/\sqrt{n}} < \frac{c_1 + z_\alpha \cdot \frac{\sigma}{\sqrt{n}} - c_2}{\sigma/\sqrt{n}}\right)$$

Therefore,
$$\beta = P\left(Z \leq z_\alpha + \frac{c_1 - c_2}{\sigma/\sqrt{n}}\right)$$

Example

Suppose we want to test whether a new teaching method has increased student performance. The test scores are normally distributed with a known standard deviation $\sigma = 10$. A random sample of size $n = 25$ is taken.

We want to test:

$$H_0 : \mu = 70 \quad \text{vs} \quad H_1 : \mu = 75$$

Rule: Reject H_0 in favor of H_1 if the sample mean $\bar{X} \geq 73.29$

Z-Test for Mean (Known σ)

Define the test statistic:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{\bar{X} - 70}{2}$$

Rejection rule: Reject H_0 if $\bar{X} \geq 73.29$, which corresponds to:

$$Z \geq \frac{73.29 - 70}{2} = 1.645$$

Error Probabilities

- **Type I Error:** Rejecting H_0 when $\mu = 70$

$$P(\bar{X} \geq 73.29 \mid \mu = 70) = P(Z \geq 1.645) = 0.05$$

- **Type II Error:** Failing to reject H_0 when $\mu = 75$

$$P(\bar{X} < 73.29 \mid \mu = 75) = P\left(Z < \frac{73.29 - 75}{2}\right) = P(Z < -0.855) = 0.196$$

Hypotheses: $H_0 : \mu = 70$ vs $H_1 : \mu = 75$

Sample size: $n = 25$, $\sigma = 10$

Rule $\bar{X} \geq$	Type I Error	Type II Error
73.29	0.05	0.196
74.00	0.0062	0.370
72.00	0.1587	0.0918
71.00	0.3085	0.0446
70.00	0.5	0.0062

Effect of Sample Size with Various Critical Values

Hypotheses: $H_0 : \mu = 70$ vs $H_1 : \mu = 75$

Population standard deviation: $\sigma = 10$

Sample Size n	Critical $\bar{X}$	Type I Error	Type II Error
25	73.29	0.050	0.196
25	74.00	0.0062	0.370
25	72.00	0.1587	0.0918
100	71.65	0.050	0.00003
100	72.00	0.0228	0.0004
100	73.00	0.0013	0.0082

Critical value is calculated using $\bar{X}_{\text{crit}} = \mu_0 + z_{\alpha} \cdot \frac{\sigma}{\sqrt{n}}$, where $z_{0.05} = 1.645$

Test for Sample Mean when Population Variance is Unknown.

Formal Setup

Suppose $X \sim N(\mu, \sigma^2)$ where $\mu \rightarrow$ unknown, $\sigma \rightarrow$ unknown.
 $X_1, X_2, \dots, X_n$ is an i.i.d. sample from population X which is known.

$$H_0 : \mu = c_1 \quad \text{vs} \quad H_1 : \mu = c_2 \quad \text{with } c_2 > c_1$$

Rule: H_0 is rejected in favour of H_1 if $\bar{X} \geq c_1 + t_{\alpha, n-1} \cdot \frac{s}{\sqrt{n}}$

Test Statistic: $T = \frac{\bar{X} - c_1}{s/\sqrt{n}} \sim t_{n-1}$ under H_0

Note:

σ is known.

$X \sim \mathcal{N}(\mu, \sigma^2)$.

Probability of Type I Error

$$\alpha = P(\text{Type I error}) = P(\text{Reject } H_0 \mid \mu = c_1) = P\left(\bar{X} \geq c_1 + t_{\alpha, n-1} \cdot \frac{S}{\sqrt{n}} \mid \mu = c_1\right)$$

Under H_0 , we have: $T = \frac{\bar{X} - c_1}{S/\sqrt{n}} \sim t_{n-1}$. Therefore, $\alpha = P(T \geq t_{\alpha, n-1})$.

Note: α is usually chosen by the researcher (e.g. 0.05 or 0.01).

Probability of Type II Error

$$\beta = P(\text{Type II error}) = P(\text{Failing to reject } H_0 \mid \mu = c_2) = P\left(\bar{X} < c_1 + t_{\alpha, n-1} \cdot \frac{S}{\sqrt{n}} \mid \mu = c_2\right)$$

Under H_1 , we have $T = \frac{\bar{X} - c_2}{S/\sqrt{n}} \sim t_{n-1}$

$$\beta = P\left(\bar{X} < c_1 + t_{\alpha, n-1} \cdot \frac{S}{\sqrt{n}} \mid \mu = c_2\right) = P\left(\frac{\bar{X} - c_2}{S/\sqrt{n}} < \frac{c_1 + t_{\alpha, n-1} \cdot \frac{S}{\sqrt{n}} - c_2}{S/\sqrt{n}}\right)$$

Therefore, $\beta = P\left(T \leq t_{\alpha, n-1} + \frac{c_1 - c_2}{S/\sqrt{n}}\right)$

Example

Suppose a pharmaceutical company is testing whether a new drug reduces systolic blood pressure more than the current standard.

The systolic blood pressure reductions (in mmHg) are assumed to be normally distributed, the sample standard deviation is known to be 5mmHg but the population standard deviation is unknown.

A sample of $n = 25$ patients is taken.

We want to test:

$$H_0 : \mu = 10 \quad \text{vs} \quad H_1 : \mu = 15$$

Rule: Reject H_0 in favor of H_1 if the sample mean $\bar{X} \geq 13.29$

t-Test for Mean (Unknown σ)

Since the population standard deviation is unknown, we use:

$$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}$$

For $n = 25$, degrees of freedom = 24. At $\alpha = 0.05$, one-sided test:

$$t_{\alpha, n-1} = 1.711 \Rightarrow \text{Reject } H_0 \text{ if } t \geq 1.711$$

Assuming sample standard deviation $s = 5$, then:

$$\bar{X}_{\text{crit}} = 10 + 1.711 \cdot \frac{5}{\sqrt{25}} = 10 + 1.711 = 11.711$$

Error Probabilities (Assuming $s = 5$)

- Type I Error:

$$P(t \geq 1.711 \mid \mu = 10) = 0.05$$

- Type II Error:

$$P(t < 1.711 \mid \mu = 15)$$

When $\mu = 15$, mean shift:

$$t = \frac{11.711 - 15}{5/\sqrt{25}} = \frac{-3.289}{1} = -3.289 \Rightarrow P(t < -3.289) \approx 0.0006$$

Hypotheses: $H_0 : \mu = 10$ vs $H_1 : \mu = 15$

Sample size: $n = 25, s = 5$, Degrees of freedom = 24

Rule $\bar{X} \geq$	Type I Error	Type II Error
11.71	0.050	0.0006
12.00	0.025	0.0019
13.00	0.002	0.056
14.00	0.00007	0.308
15.00	0.000003	0.5

Effect of Sample Size with Various Critical Values

Hypotheses: $H_0 : \mu = 10$ vs $H_1 : \mu = 15$

Assume $s = 5$, Use appropriate t_{crit} for each n

Sample Size n	Critical $\bar{X}$	Type I Error	Type II Error
25	11.71	0.050	0.0006
50	11.40	0.050	0.0000005
75	11.29	0.050	$< 10^{-9}$
100	11.23	0.050	$< 10^{-12}$

Critical value: $\bar{X}_{\text{crit}} = \mu_0 + t_{\alpha} \cdot \frac{s}{\sqrt{n}}$

Test for Sample Mean when Population Variance is Known.

Formal Setup

Suppose $X \sim N(\mu, \sigma^2)$ where $\mu \rightarrow$ unknown, $\sigma \rightarrow$ known.
 $X_1, X_2, \dots, X_n$ is an i.i.d. sample from population X which is known.
We want to test: $H_0 : \mu = c$ versus $H_1 : \mu > c$.

Question

If the null hypothesis is true, how likely is it that we would have seen a sample mean as large as $\bar{X}$?

Test for Sample Mean when Population Variance is Known.

Answer

Assume that empirical values of $X_1, X_2, \dots, X_n$ are known.
Let $Y_1, Y_2, \dots, Y_n$ be any i.i.d. sample from population X .

Note:

$$X \sim N(\mu, \sigma^2)$$

$\bar{Y} = \frac{Y_1 + Y_2 + \dots + Y_n}{n}$ is a random variable.

$\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n}$ is the deterministic observed sample average.

Test for Sample Mean when Population Variance is Known.

Answer (Continued)

If the null hypothesis is true, $Z = \frac{\sqrt{n}(\bar{Y} - c)}{\sigma} \sim N(0, 1)$.

$$P(\bar{Y} \geq \bar{X}) = P\left(\frac{\sqrt{n}(\bar{Y} - c)}{\sigma} \geq \frac{\sqrt{n}(\bar{X} - c)}{\sigma}\right) = P\left(Z \geq \frac{\sqrt{n}(\bar{X} - c)}{\sigma}\right)$$

Note:

In practice, before a sample is taken, a *level of significance* $\alpha \in (0, 1)$ is typically fixed.

Definition of p-value

General Idea

The **p-value** is the probability, assuming the null hypothesis H_0 is true, of observing a test statistic at least as extreme as the one calculated from the sample.

In Our Current Example

$\bar{X}$ is the observed sample mean.

$$\text{p-value} = P(\bar{Y} \geq \bar{X} | H_0) = P\left(Z \geq \frac{\sqrt{n}(\bar{X} - c)}{\sigma}\right)$$

Decision:

- If $P(\bar{Y} \geq \bar{X}) < \alpha$ then the sample average is far away from assumed mean c .
 $H_0 : \mu = c$ is judged to be incorrect. **We reject the null hypothesis.**
- If $P(\bar{Y} \geq \bar{X}) \geq \alpha$ then the sample average seems to be consistent with the assumption $\mu = c$. **We do not reject the null hypothesis.**

Example

- Toss a coin 10 times.
- Test: $H_0 : p = 0.5$ vs. $H_1 : p = 0.6$
- **Rule:** Reject H_0 if ≥ 7 heads observed.

p-value = $P(X \geq 7 \mid p = 0.5)$ where $X \sim \text{Binomial}(10, 0.5)$

$$P(X \geq 7 \mid p = 0.5) = \sum_{k=7}^{10} \binom{10}{k} (0.5)^{10} = \frac{11}{64} \approx \boxed{0.1719}$$

References

- 1 Fundamentals of Mathematical Statistics (tenth revised edition), by S.C. Gupta and V.K. Kapoor, Sultan Chand & Sons.
- 2 Probability and Statistics with Examples using R, by Siva Athreya, Deepayan Sarkar, and Steve Tanner, <https://psweur.github.io> (2024).

Thank you

